

ORF307

Duality II

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Today's topics

- Two-person zero-sum games
- Farkas lemma
- Complementarity slackness
- KKT conditions (Karush-Kuhn-Tacker)

- **Recap**

Optimal objective values

Primal

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax \leq b\end{array}$$

p^* is the primal optimal value

Infeasible Primal $\rightarrow p^* = +\infty$

Unbounded Primal $\rightarrow p^* = -\infty$

Dual

$$\begin{array}{ll}\max_{y} & -b^T y \\ \text{s.t.} & A^T y + c = 0 \\ & y \geq 0\end{array}$$

d^* is the primal optimal value

Infeasible Dual $\rightarrow d^* = -\infty$

Unbounded Dual $\rightarrow d^* = +\infty$

Weak duality

Theorem: if $x \in R^n$ and $y \in R^m$ satisfy

$$\left. \begin{array}{l} \bullet \ x \text{ is a feasible solution to the Primal} \\ \bullet \ y \text{ is a feasible solution to the Dual} \end{array} \right\} \Rightarrow -b^T y \leq c^T x$$

Proof:

$$\left. \begin{array}{l} \text{Since } x \text{ is feasible for the Primal: } Ax \leq b \Rightarrow y^T Ax \leq y^T b. \\ \text{Since } y \text{ is feasible for the Dual: } Ay + c = 0 \text{ and } y \geq 0. \end{array} \right\} \Rightarrow y^T Ax \leq y^T b \Rightarrow -c^T x \leq b^T y \Rightarrow -b^T y \leq c^T x$$


Remarks:

- For any dual feasible solution y , the dual objective $-b^T y$ is a **lower bound** to the primal objective $c^T x$
- For any primal feasible solution x , the primal objective $c^T x$ is an **upper bound** to the dual objective $-b^T y$
- **Duality gap:** $g = c^T x + b^T y$

Week duality – Corollaries

Unboundedness –vs- feasibility

- Primal unboundedness ($p^* = -\infty$) $\Rightarrow$ Dual infeasibility ($d^* = -\infty$)
- Dual unboundedness ($d^* = +\infty$) $\Rightarrow$ Primal infeasibility ($p^* = +\infty$)

Optimality conditions

IF x, y satisfy:

- x is feasible solution to the primal LP
- y is feasible solution to the dual LP
- The duality gap is zero (i.e., $g = c^T x + b^T y = 0$)

THEN x is an optimal solution to the Primal and y is an optimal solution to the Dual.

Strong duality

Theorem:

If an LP (the primal) has an optimal solution x^* then its dual LP has also an optimal solution y^* , and the optimal objective values of the Primal and Dual LPs are equal, i.e.,

$$\mathbf{p}^* = c^T x^* = -b^T y^* = \mathbf{d}^*$$

In such a case the duality gap is zero.

Proof: see last lecture slides

Relationships between Primal and Dual problems

Primal \ Dual	$p^* = +\infty$ (infeasible)	p^* finite (optimal)	$p^* = -\infty$ (unbounded)
$d^* = +\infty$ (unbounded)	primal inf. dual unb.	Impossible	Impossible
d^* finite (optimal)	Impossible	optimal values equal	Impossible
$d^* = -\infty$ (infeasible)	exception	Impossible	primal unb. dual inf

- Upper-right excluded by weak duality
- (1,1) and (3,3) are proven using weak duality
- (3,1) and (2,2) are proven using strong duality

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- **2-Person-0-Sum games**

Rock or Paper or Scissors

Rules of the game:

At count to three, the two players say one of the following: **Rock** or **Paper** or **Scissors**

Winners:

- **Rock** beats ("dulls") **scissors**
- **Scissors** beats ("cuts") **paper**
- **Paper** beats ("covers") **rock**

Identical selection is a **draw**

Questions:

1. Do you always choose the same object?
Who has the advantage?
2. Do you randomly select an object?
Who has the advantage?

Extremely popular game: world RPS society, USA RPS league, etc.

2-Person-0-Sum games

- **Player 1 (P1)** chooses a number $i \in \{1, 2, \dots, m\}$ (one of m actions)
- **Player 2 (P2)** chooses a number $j \in \{1, 2, \dots, n\}$ (one in n actions)
- The two players make their choices **independently**

Rule:

Player 1 pays A_{ij} to Player 2

$A \in R^{m \times n}$ is the **payoff matrix**

Payoff matrix for RPS game

$$A = \begin{array}{ccc|c} & R & P & S \\ \hline R & 0 & 1 & -1 \\ P & -1 & 0 & 1 \\ S & 1 & -1 & 0 \end{array}$$

Mixed (randomized) strategies

Deterministic strategies can be systematically defeated

Randomized strategies

- **P1** chooses **randomly** according to **distribution** x
 - x_i = **probability** that P1 selects action $i \in \{1, 2, \dots, m\}$
- **P2** chooses **randomly** according to **distribution** y
 - y_j = **probability** that P2 selects action $j \in \{1, 2, \dots, n\}$
- Expected payoff (from P1 to P2), if they use mixed-strategies x and y

$$\sum_{i=1}^m \sum_{j=1}^n x_i A_{ij} y_j = x^T A y$$

Mixed strategies and probability simplex

A **probability simplex** in R^k is defined as the **set**:

$$P_k = \{p \in R^k \mid \mathbf{1}^T p = 1 \text{ and } p \geq 0\}$$

Example: in R^3 the probability simplex is defined as

$$P_3 = \{p \in R^3 \mid p_1 + p_2 + p_3 = 1 \text{ and } p_1, p_2, p_3 \geq 0\}$$

Mixed strategy

For a game player, mixed strategy is a **distribution** over all possible deterministic strategies

The set of all mixed strategies is the probability simplex

- $x \in P_m$
- $y \in P_n$

Recall the 3 Axioms from Probability Theory (ORF309)

Optimal mixed strategies

For player P1: optimal strategy x^* is the solution of

Minmax problem:

$$\begin{aligned} \min_{x \in P_m} \max_{y \in P_n} x^T A y &\Rightarrow \min_{x \in P_m} \max_{j=1, \dots, n} (A^T x)_j \end{aligned}$$

P1 is choosing a mixed strategy x to **minimize the maximum possible payoff** (i.e., the expected value of the game) that **P2** can force.

Expected \$\$ that **P1** needs to pay to **P2**, if **P1** plays x and **P2** chooses deterministic strategy j . **P2** wants to maximize the \$\$ he receives and **P1** wants to minimize that maximum amount of \$\$

For player P2: optimal strategy y^* is the solution of

Maxmin problem:

$$\begin{aligned} \max_{y \in P_n} \min_{x \in P_m} x^T A y &\Rightarrow \max_{y \in P_n} \min_{i=1, \dots, m} (A y)_i \end{aligned}$$

P2 is choosing a mixed strategy y to **maximize the minimum possible payoff** (i.e., the expected value of the game) that **P1** can force upon them.

Inner optimization problems is over **deterministic** strategies (which are vertices of P_m or P_n)

The **optimal** mixed strategies x^* and y^* are computed using **Linear Optimization**

Minimax theorem (for 2-Person-0-Sum games only)

Theorem (John von Neumann, 1928):

$$\max_{y \in P_n} \min_{x \in P_m} x^T A y = \min_{x \in P_m} \max_{y \in P_n} x^T A y = V$$

Proof:

- V is the **value of the game**
- **Significance:** defines the optimal outcome and strategies for pure competitive games.

The optimal x^* is the solution of

$$\begin{aligned} \min \quad & t \\ \text{s.t.} \quad & A^T x < t \mathbf{1} \\ & \mathbf{1}^T x = 1 \\ & x \geq 0 \end{aligned}$$

Always has an optimal solution

The optimal y^* is the solution of

$$\begin{aligned} \min \quad & w \\ \text{s.t.} \quad & A^T y < w \mathbf{1} \\ & \mathbf{1}^T y = 1 \\ & y \geq 0 \end{aligned}$$

Always has an optimal solution

The two LPs are **duals of each other** and by **strong duality** the equality follows



Nash equilibrium (Non-Zero-Sum Games with many players)

Theorem (John Nash, 1950): Every **general game** has at least one equilibrium in mixed strategies.

Definition: A Nash equilibrium is a set of strategies (one of each player), such that no player can improve their payoff by changing their strategy when the other players do not change theirs.

- **Remark:** Connection between von Neumann theorem (Minimax theorem) and a Nash equilibrium.
 - Nash's theorem is a **generalization** of von Neumann's theorem.
 - For **2-Player-0-Sum games** the **mixed strategies** (x^*, y^*) defined by the **Minimax theorem** are also a **Nash equilibrium**

$$x^T A y^* \geq x^{*T} A y^* \geq x^{*T} A y, \quad \forall x \in P_m, \quad \forall y \in P_n$$

Nash equilibrium of two-person zero-sum game,

Example

A_{ij} : payoff from **P1** to **P2**

$$A = \begin{matrix} & \text{P2 (columns)} \\ \text{P1 (rows)} & \begin{bmatrix} 4 & 2 & 0 & -3 \\ -2 & -4 & -3 & 3 \\ -2 & -3 & 4 & 1 \end{bmatrix} \end{matrix}$$

Optimal deterministic strategies

$$\mathbf{3} = \min\{4, 3, 4\} = \min_i \max_j A_{ij} \geq \max_j \min_i A_{ij} = \max\{-2, -4, -3\} = -2$$

No equilibrium → if a **player commits** to a strategy, the **other player** who moves second **can exploit it!**

Optimal mixed strategies (solving the LPs)

$$\mathbf{x}^* = (0.37, 0.33, 0.3)^T, \quad \mathbf{y}^* = (0.4, 0.13, 0.47)^T$$

Probabilities of selecting each strategy by the players

Expected payoff:

$$\mathbf{x}^{*T} A \mathbf{y}^* = 0.2$$

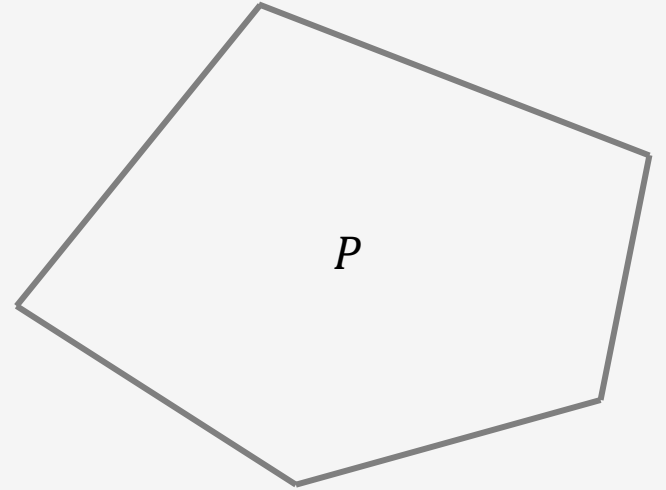
Non-zero-sum game – Example

- **The prisoner's dilemma (classic example!):**
- Two **usual thieves** have been caught and are interrogated separately
- Each suspect has two choices (strategies):
 - Stay silent (cooperate with the other thief) or testify against his friend (defect).
 - If both cooperate, both get 1 year in prison.
 - If one defects and the other cooperates,
 - The defector goes free (0 years in prison)
 - The cooperator gets 10 years in prison (Alcatraz) .
 - If both defect (they testify against each other),
 - Both get 5 years in prison.
 - The payoffs do not sum to zero in any case (can you write the payoff matrix?)
 - If both stay silent they get a lower sentence (thieves have to be loyal to each other...)

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- **Farkas lemma**

Feasibility of polyhedra

$$P = \{x \mid Ax = b, \quad x \geq 0\}$$



Question: How to show that P is **feasible**?

Answer: we simply need to find a point $x \in P$

x serves as a certificate of feasibility

Question: How to show that P is infeasible?

This has a **more complicated answer** → **Farkas lemma**

Farkas lemma

$$P = \{x \mid Ax = b, \quad x \geq 0\}$$

Theorem:

Let $A \in R^{m \times n}$ and $b \in R^m$. Then **exactly one** of the following statements is **True**:

1. There exists an $x \in R^n$ such that $Ax = b$ and $x \geq 0$. That is:

$$b = \sum_{i=1}^n x_i A_i, \quad \text{and} \quad x_i \geq 0, i = 1, \dots, n$$

and b is **in** the **cone** K generated by the columns of A

OR (exclusive)

2. There exists a $y \in R^m$ such that $A^T y \geq 0$ and $b^T y < 0$. That is,

$$y^T A_i \geq 0, i = 1, \dots, m, \quad y^T b < 0$$

The hyperplane $y^T z = 0$, **separates** b from $A_1, \dots, A_n$

Farkas lemma

Let's define what a **Cone** is:

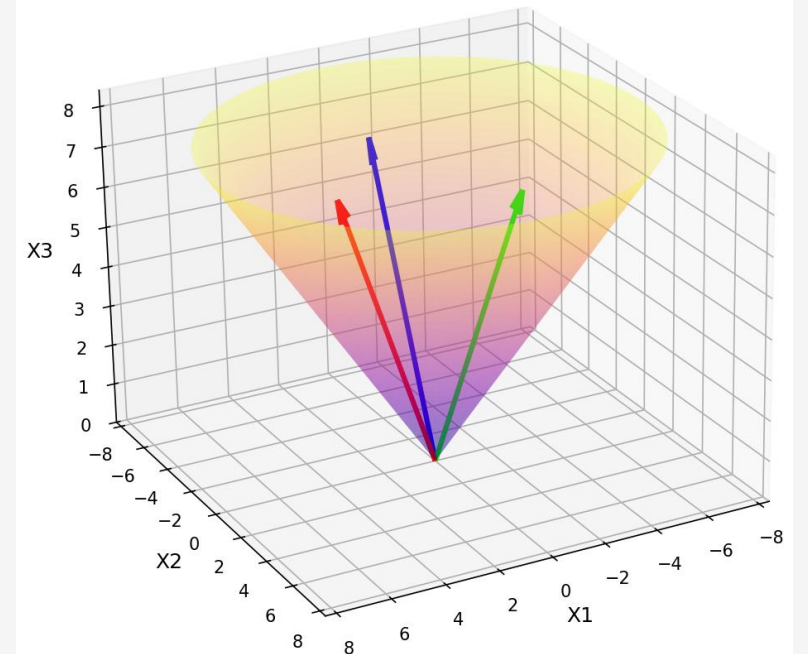
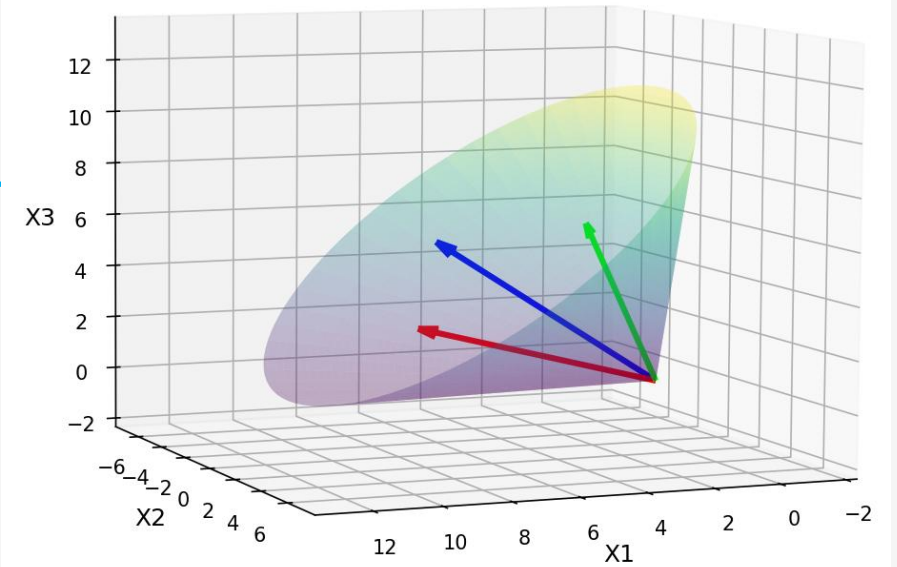
A (convex) **cone** generated by n m -dimensional vectors $a_1, \dots, a_n \in R^m$ is defined by the set

$$K = \{a_1x_1 + \dots + a_nx_n \mid x_1, \dots, x_n \geq 0\}$$

The cone can be equivalently written as

$$K = \{Ax \mid x \geq 0\}$$

where we have placed the generating vectors $a_i, i = 1, \dots, n$ as columns of the matrix $A \in R^{m \times n}$



Farkas lemma

Geometric interpretation (in 2D):

First alternative

There exists an x with $Ax = b, x \geq 0$:

$$b = \sum_{i=1}^n x_i A_i, \quad x_i \geq 0, i = 1, \dots, n$$

b is **in** the **cone** K generated by the columns of A

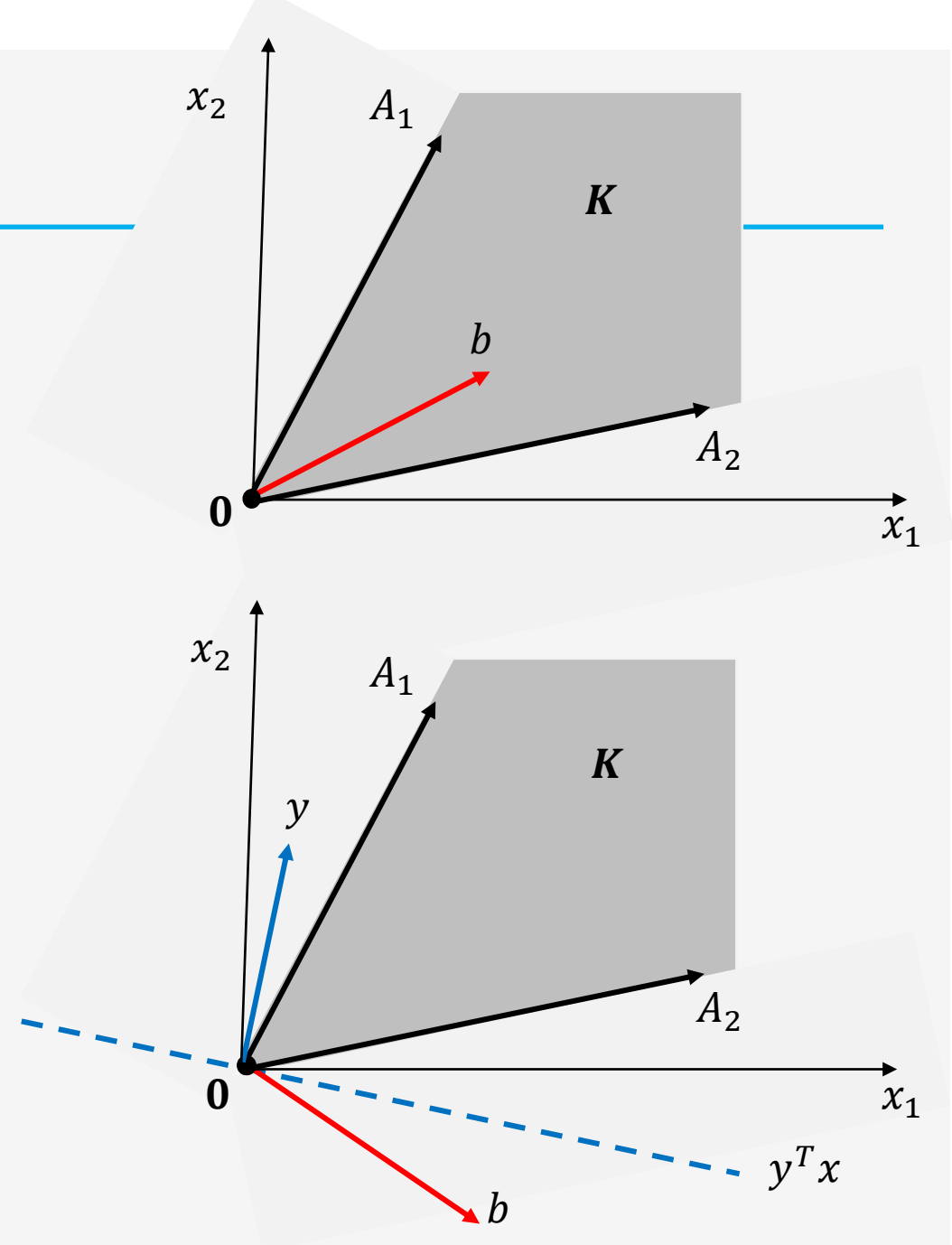
OR (exclusive)

Second alternative

There exists y with $A^T y \geq 0, b^T y < 0$

$$y^T A_i \geq 0, i = 1, \dots, m, \quad y^T b < 0$$

The hyperplane $y^T z = 0$, **separates** b from $A_1, \dots, A_n$



Farkas lemma

(1) There exists an x with $Ax = b, x \geq 0$ **OR** (2) There exists a y with $A^T y \geq 0, b^T y < 0$

Proof

1 and 2 cannot be true at the same time (easy part)

Assume that: $x \geq 0, Ax = b$ **and** $y^T A \geq 0, b^T y < 0 \Rightarrow y^T b = y^T Ax \geq 0$ **Contradiction**

$\underbrace{x \geq 0, Ax = b}_{\text{Option 1 holds}} \quad \text{and} \quad \underbrace{y^T A \geq 0, b^T y < 0}_{\text{Option 2 holds}} \Rightarrow \underbrace{y^T b = y^T Ax \geq 0}_{\text{Contradiction}}$

Multiply by y^T the equations $Ax = b$ and use the 1st condition of Option 2 to derive $y^T Ax \geq 0$ which **contradicts** with the 2nd condition of Option 2.

Farkas lemma

(1) There exists an x with $Ax = b, x \geq 0$ **OR** (2) There exists a y with $A^T y \geq 0, b^T y < 0$

Proof

1 and 2 cannot be false at the same time (duality)

Primal: (P)

min 0

s.t. $Ax = b$

$x \geq 0$

Observation:

If (P) is **feasible** is
not unbounded,
because its objective
is **constant**

Dual: (D)

max $-b^T y$

s.t. $A^T y \geq 0$

Observation:

$y = 0$ is **always**
feasible and makes
the objective 0



From **strong duality** we have:

Dual is feasible ($d^* \neq -\infty$) and $p^* = d^*$,
where p^*, d^* are the optimal objective
values of the Primal and Dual problems.

Farkas lemma

(1) There exists an x with $Ax = b, x \geq 0$ **OR** (2) There exists a y with $A^T y \geq 0, b^T y < 0$

Proof

1 and 2 cannot be false at the same time (duality)

Primal: (P)

$\min 0$
s.t. $Ax = b$
 $x \geq 0$

Observation:

If (P) is **feasible** is **not unbounded**, because its objective is **constant**

Dual: (D)

$\max -b^T y$
s.t. $A^T y \geq 0$

Observation:

$y = 0$ is **always feasible** and makes the objective 0

Strong duality:

$$d^* \neq -\infty, \quad p^* = d^*$$

Alternative 1: The primal is **feasible** $\Rightarrow$

$$\Rightarrow p^* = d^* = 0 \Rightarrow -b^T y \leq 0 \Rightarrow b^T y \geq 0$$

So, we showed that: $b^T y \geq 0$, for all $y : A^T y \geq 0$

Hence: (2) is false!

Farkas lemma

(1) There exists an x with $Ax = b, x \geq 0$ **OR** (2) There exists a y with $A^T y \geq 0, b^T y < 0$

Proof

1 and 2 cannot be false at the same time (duality)

Primal: (P)

$\min 0$

s.t. $Ax = b$

$x \geq 0$

Observation:

If (P) is **feasible** is **not unbounded**, because its objective is **constant**

Dual: (D)

$\max -b^T y$

s.t. $A^T y \geq 0$

Observation:

$y = 0$ is **always feasible** and makes the objective 0

Strong duality:

$$d^* \neq -\infty, \quad p^* = d^*$$

Alternative 2: The (P) is **infeasible** $\Rightarrow$ (D) is **UNBND**

Then $\exists$ sequence of y^k 's: $A^T y^k \geq 0$ & $-b^T y^k \rightarrow +\infty$

$\Rightarrow \exists y: A^T y \geq 0$ & $-b^T y > 0 \Rightarrow \underbrace{\exists y: A^T y \geq 0 \text{ \& } b^T y < 0}_{\text{Option 2 is True}}$

**Option 2
is True**

Farkas lemma – many variations

Variation #1:

(1) There exists an x with $Ax = b, x \geq 0$ **OR** (2) There exists a y with $A^T y \geq 0, b^T y < 0$

Variation #2:

(1) There exists an x with $Ax \leq b, x \geq 0$ **OR** (2) There exists a y with $A^T y \geq 0, b^T y < 0, y \geq 0$

Variation #3:

(1) There exists an x with $Ax \leq b$ **OR** (2) There exists a y with $A^T y = 0, b^T y < 0, y \geq 0$

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- **Complementarity slackness**

Optimality conditions

Primal:

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax \leq b\end{array}$$

Dual:

$$\begin{array}{ll}\max & -b^T y \\ \text{s.t.} & A^T y + c = 0 \\ & y \geq 0\end{array}$$

x and y are primal and dual optimal if and only if (iff)

- x is **primal feasible**: $Ax \leq b$
- y is **dual feasible**: $A^T y + c = 0$, $y \geq 0$
- The **duality gap is zero**: $c^T x + b^T y = 0$

Can we find a relationship between x and y directly?

(not only as they relate in the objectives)

Complementarity slackness

Primal:

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax \leq b\end{array}$$

Dual:

$$\begin{array}{ll}\max & -b^T y \\ \text{s.t.} & A^T y + c = 0 \\ & y \geq 0\end{array}$$

Theorem:

Let x be primal feasible and y be dual feasible. Then x, y are optimal if and only if (iff)

$$y_i(b - a_i^T x) = 0, \quad i = 1, 2, \dots, m$$

That is, **at an optimum** (primal or dual), $b - Ax$ and y have **complementary sparsity pattern**:

$$y_i > 0 \Rightarrow a_i^T x - b = 0$$

$$a_i^T x < b_i \Rightarrow y_i = 0$$

Complementarity slackness

Primal:

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax \leq b\end{array}$$

Dual:

$$\begin{array}{ll}\max & -b^T y \\ \text{s.t.} & A^T y + c = 0 \\ & y \geq 0\end{array}$$

Proof:

The duality gap at primal feasible x and dual feasible y can be written as:

$$c^T x + b^T y = (-A^T y)^T x + b^T y = (b - Ax)^T y = \sum_{i=1}^m y_i (b_i - a_i^T x) = 0$$

Note that the elements of the sum are all nonnegative and the sum must be zero.

Hence all the elements must be zero. 

Remark: for feasible x and y , we have $c^T x + b^T y = 0 \Leftrightarrow y_i (b_i - a_i^T x) = 0, i = 1, \dots, m$

Example

Usefulness: Complementarity helps us recover the dual solution when we know a primal solution

Consider the LP:

$$\begin{array}{ll} \min & -4x_1 - 5x_2 \\ \text{s.t.} & \begin{bmatrix} -1 & 0 \\ 2 & 1 \\ 0 & -1 \\ 1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 0 \\ 3 \\ 0 \\ 3 \end{pmatrix} \end{array}$$

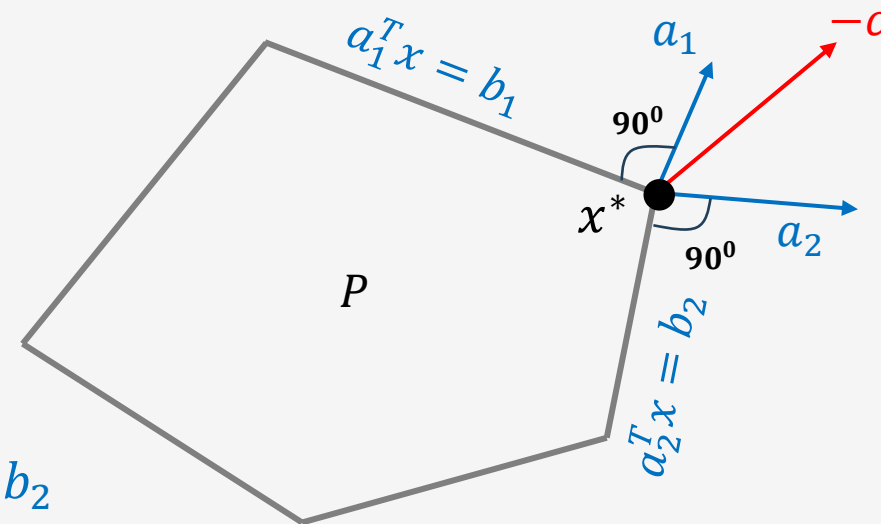
Let's show that the feasible point $x = (1,1)^T$ is optimal

The 2nd and 4th constraints are **active (tight)** at x which implies: $y = (0, y_2, 0, y_4)^T$ (because of **complementarity**)

From the dual constraints we have: $A^T y = -c \Rightarrow \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{pmatrix} y_2 \\ y_4 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$, and $y_2, y_4 \geq 0$

Solving the linear system $\rightarrow y = (0, 1, 0, 2)$ satisfies these conditions and proves that x is optimal

Geometric interpretation



Two **active constraints** at the **optimum**: $a_1^T x^* = b_1$ and $a_2^T x^* = b_2$

The **dual optimal solution** y satisfies:

$$A^T y + c = 0, \quad y \geq 0, \quad \text{and} \quad y_i = 0, \quad \text{for } i \neq 1, 2$$

Hence: $A^T y + c = 0 \Rightarrow a_1 y_1 + a_2 y_2 = -c$ **since only** $y_1, y_2 \geq 0$

Note: a_i^T is the i -th row of A and for A^T we have a_i as the i -th column

$$A = \begin{bmatrix} a_1^T \\ \vdots \\ a_m^T \end{bmatrix}$$

$$A^T = [a_1 \quad \cdots \quad a_m]$$

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- **KKT conditions**

Lagrangian and duality

Primal:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \end{aligned}$$

Dual:

$$\begin{aligned} \max \quad & -b^T y \\ \text{s.t.} \quad & A^T y + c = 0 \\ & y \geq 0 \end{aligned}$$

Dual function:

$$\begin{aligned} g(y) &= \min_x (c^T x + y^T (Ax - b)) \\ &= -b^T y + \min_x (c + A^T y)^T x = \\ &= \begin{cases} -b^T y, & \text{if } c + A^T y = 0 \\ -\infty, & \text{otherwise} \end{cases} \end{aligned}$$

Lagrangian function

$$L(x, y) = c^T x + y^T (Ax - b)$$

NOTE: $\nabla_x L(x, y) = 0 \Leftrightarrow c + A^T y = 0$

KKT conditions

Optimality conditions of LP

Primal:

$$\begin{array}{ll} \min & c^T x \\ \text{s.t.} & Ax \leq b \end{array}$$

Dual:

$$\begin{array}{ll} \max & -b^T y \\ \text{s.t.} & A^T y + c = 0 \\ & y \geq 0 \end{array}$$

Primal feasibility: $Ax \leq b$

Dual feasibility: $A^T y + c = 0$, $y \geq 0$

Complementarity slackness: $y_i(Ax - b)_i = 0, i = 1, \dots, m$

KKT conditions

Optimality conditions of LP

Primal:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \end{aligned}$$

Dual:

$$\begin{aligned} \max \quad & -b^T y \\ \text{s.t.} \quad & A^T y + c = 0 \\ & y \geq 0 \end{aligned}$$

We can solve our LP by solving the **system of linear equations and inequalities**

$$\nabla_x L(x, y) = A^T y + c = 0$$

$$y \geq 0$$

$$b - Ax \geq 0$$

$$y^T (b - Ax) = 0$$

references

- B/T
 - Ch. 4: duality
- V
 - Game theory

Next lecture

- Sensitivity
